

I checked current public datasets, and there is a good way to build the project around **battery degradation + UAV operating conditions + altitude/environmental factors**.

Best dataset combination

1. NASA Li-ion Battery Aging Dataset — primary dataset

NASA's dataset contains battery charge/discharge cycles at different temperatures, with **voltage, current, battery temperature, ambient temperature and capacity** measurements. It was specifically collected for prognostics and Remaining Useful Life (RUL) prediction.

[NASA Li-ion Battery Aging Dataset](#)

2. NASA Randomized/Recommissioned Battery Dataset — additional validation

This contains 26 battery packs tested under different constant and variable loading conditions and includes degradation, temperature, voltage, current and capacity information.

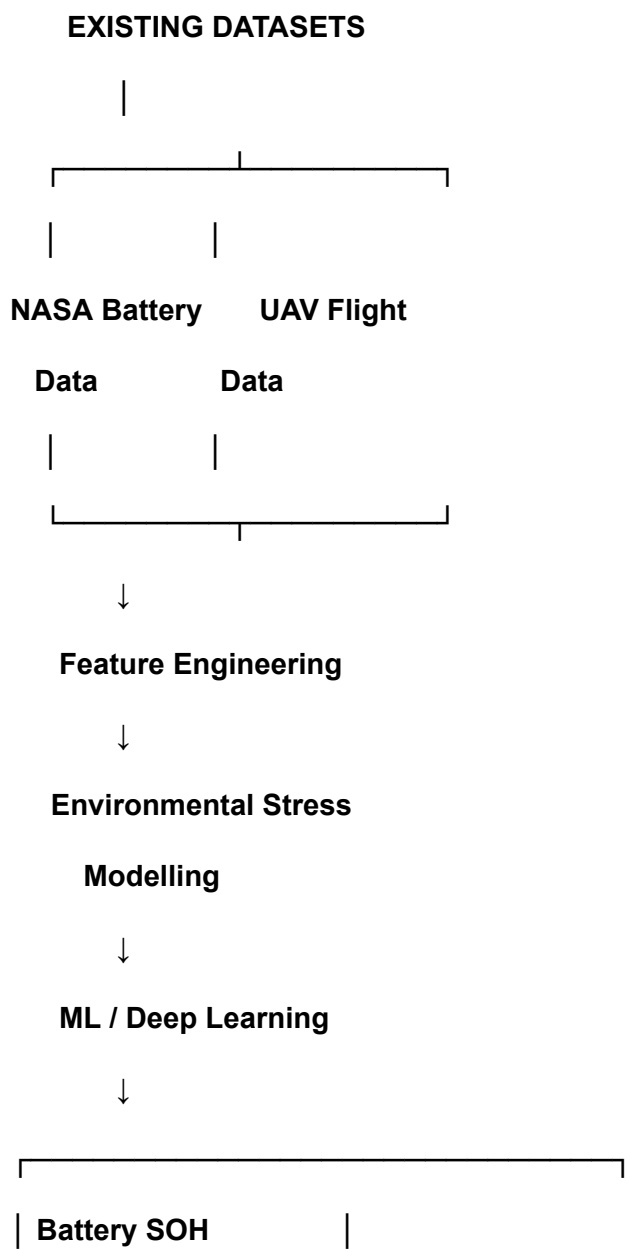
[NASA Randomized and Recommissioned Battery Dataset](#)

3. AMOVFLY UAV dataset — very interesting for your hackathon

This dataset includes UAV flight information such as **flight duration, power consumption, battery number, atmospheric pressure, air density, temperature, wind speed and weather**. That makes it particularly useful for connecting UAV performance with environmental conditions.

[AMOVFLY Dataset](#)

AI-Based High-Altitude Equipment Reliability and Remaining Useful Life Prediction System

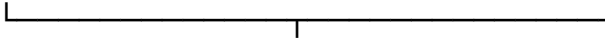


| Remaining Useful Life |

| Failure Risk |

| Energy Consumption |

| Flight Endurance |



Adaptive Recommendation



"Continue / Reduce Load /

Pre-heat Battery / Replace"

The key research question

Rather than simply predicting battery life, your model can investigate:

How do environmental and operating conditions affect the reliability and useful life of electrical/electronic equipment in high-altitude environments?

Then you can create an **Altitude-Environment Stress Model**.

For example:

$$S = f(T, P, \rho, I, V, Cycle)$$

where:

- T = temperature
- P = atmospheric pressure
- ρ = air density
- I = electrical load/current
- V = voltage
- Cycle = battery usage/aging

Even better: create a "Ladakh Mode"

This could be the innovative part of your software.

The user enters:

Altitude: 5,000 m

Ambient temperature: -25°C

Battery: 80% SOH

Current load: 12 A

Payload: 2 kg

Your model produces:

Environmental Stress: HIGH

Battery degradation risk: HIGH

Estimated usable capacity: XX%

Predicted remaining operational time: XX min

Recommended action: Pre-heat battery / reduce load

This directly addresses the hackathon's requirement of improving **reliability, efficiency and lifespan** without requiring you to manufacture hardware.

One important scientific point

We should **not simply claim that NASA's battery dataset is a Ladakh dataset**. It isn't. It gives us battery behaviour under controlled temperatures and loads. Likewise, the UAV dataset gives environmental/flight relationships but isn't necessarily a Ladakh-specific dataset.

So the scientifically defensible approach is:

Existing datasets → learn degradation/performance relationships → build a high-altitude environmental model → simulate Ladakh operating conditions → predict risk/performance.

That is much stronger than pretending the training data were collected in Ladakh.

I can take this much further

If you want, I can **download/examine these datasets and design the complete hackathon ML project**, including:

1. **Exact datasets to use**
2. Dataset columns
3. How to merge/use them
4. Target variable
5. Feature engineering
6. ML models to compare
7. **Ladakh simulation methodology**
8. Complete Python code

9. Graphs/results
10. Dashboard
11. System architecture
12. Innovation/novelty
13. Hackathon presentation
14. **2–3 minute pitch for judges**

The **AMOVFLY + NASA battery combination is particularly promising** because it lets us connect **UAV power/flight behaviour with temperature, pressure and air density**, which fits the hackathon statement very closely.